

Predicting student persistence/dropout via socio-academic features

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1. Introduction

Student attrition remains a persistent issue in higher education, with significant implications for students, institutions, and public education systems. In this project, I investigated whether demographic, socio-economic, and academic features can be used to predict student dropout. The goal was to develop an interpretable machine learning pipeline capable of accurately identifying students at risk while providing insights into the factors associated with student attrition and enabling the implementation of early support measures.

2. Methodology

The original target variable contained three classes: Graduate, Enrolled, and Dropout. For the purposes of this study, the task was reformulated as a binary classification problem predicting whether a student will drop out or not.

2.1 Feature engineering

The core innovation involved engineering *Completion Ratios* and *Engagement Indicators*, which provide quantitative proxies for student adaptation and effort during the critical first two semesters. Redundant or low-information features were removed, and strongly correlated variables were reduced to limit multicollinearity.

2.2 Data preprocessing pipeline

To ensure reproducibility and prevent data leakage, all preprocessing steps were integrated within a unified machine learning pipeline. The preprocessing stage addressed both missing data and feature scaling. Missing values were handled through imputation, while variables exhibiting strong skewness were transformed using a logarithmic transformation to reduce distribution asymmetry and stabilize variance. Features were subsequently standardized using either standard scaling or robust scaling. Categorical variables were encoded using one-hot encoding to convert them into a numerical representation suitable for machine learning algorithms.

2.3 Model selection

Following preprocessing, several classification algorithms were evaluated in order to identify the most suitable model for the dropout prediction task. The models considered included *Logistic Regression*, *K-Nearest Neighbors*, *Perceptron*, and *Random Forest*.

Because the dataset exhibits class imbalance, with a larger number of non-dropout instances compared to dropout cases, multiple resampling techniques were explored to improve the model's ability to correctly identify minority class observations. In particular, *Random Oversampling*, *Random Undersampling*, and the *Synthetic Minority Over-sampling Technique (SMOTE)* were tested.

Finally, dimensionality reduction techniques were also considered to investigate whether a lower-dimensional representation of the data could improve model performance or generalization. Specifically, *Principal Component Analysis (PCA)* and *Linear Discriminant Analysis (LDA)* were evaluated within the pipeline, however, these methods did not improve performance in practice and were therefore not included in the final model.

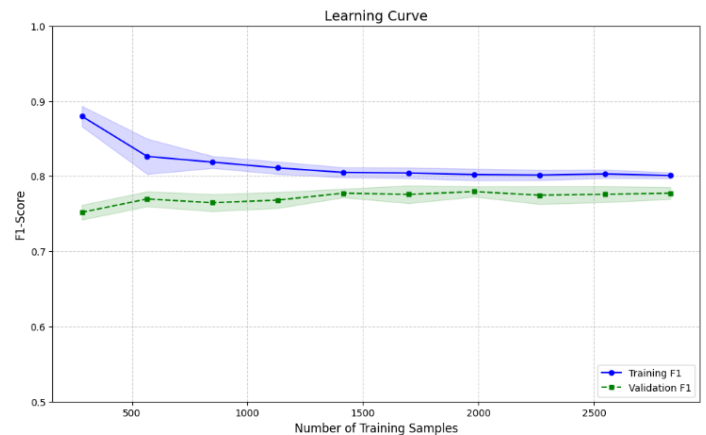
2.4 Hyperparameter optimization

Model selection and hyperparameter tuning were performed using a *nested cross-validation* framework, which provides a more reliable estimate of model generalization performance by separating the processes of model selection and evaluation. Within this framework, two hundred candidate pipeline configurations were evaluated, and each configuration was assessed in a consistent and unbiased manner.

3. Experimental Results

The experimental evaluation showed that several pipeline configurations achieved comparable levels of performance. Among them, the best-performing model consisted of a pipeline combining ***Random Oversampling*** with ***Logistic Regression*** using ***L1 regularization***. This configuration achieved an F1-score of approximately 0.81 on the held-out test set, indicating a strong balance between precision and recall when identifying students at risk of dropping out. The use of L1 regularization also provided the additional benefit of implicit feature selection, improving model interpretability.

Further analysis was conducted using a learning curve, which showed convergence between training and validation scores as the number of training samples increased. This pattern suggests that the model generalizes well and does not exhibit significant overfitting. In addition, a validation curve was used to analyze the effect of the regularization parameter C . The results indicated that optimal performance was achieved for values of C in the range $0.1 \leq C \leq 1$.



4. Conclusion

Analysis of model coefficients suggests that academic performance indicators and socio-economic background variables (such as parental education) are among the strongest predictors of dropout. In conclusion, this model can support educational institutions in designing early intervention strategies aimed at improving student retention. Future work could extend this approach by incorporating temporal data, exploring deep learning models, or applying explainability techniques to further analyze feature contributions.

Link to the project:

https://github.com/Vittoriacrugnolaa/University_projects/tree/main/Student_Dropout